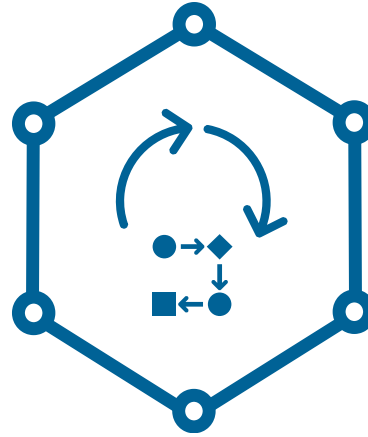




Knowledge Graph Evolution

Bonus

Emanuel Sallinger



STANFORD

Slides courtesy of Vinay K. Chaudhri
University of Stanford

Overview



Only constant in life is change

Change in the real-world
Change in the business requirements



Changes can require

Revising the schema
Revising the ground facts



Approaches to handle change
must address

Technical challenges
Social challenges

Examples Requiring Change

- Changing world
- Changing requirements
- Changing sources
- Changes affecting previous inferences
- Changes requiring redesign

Examples Requiring Change

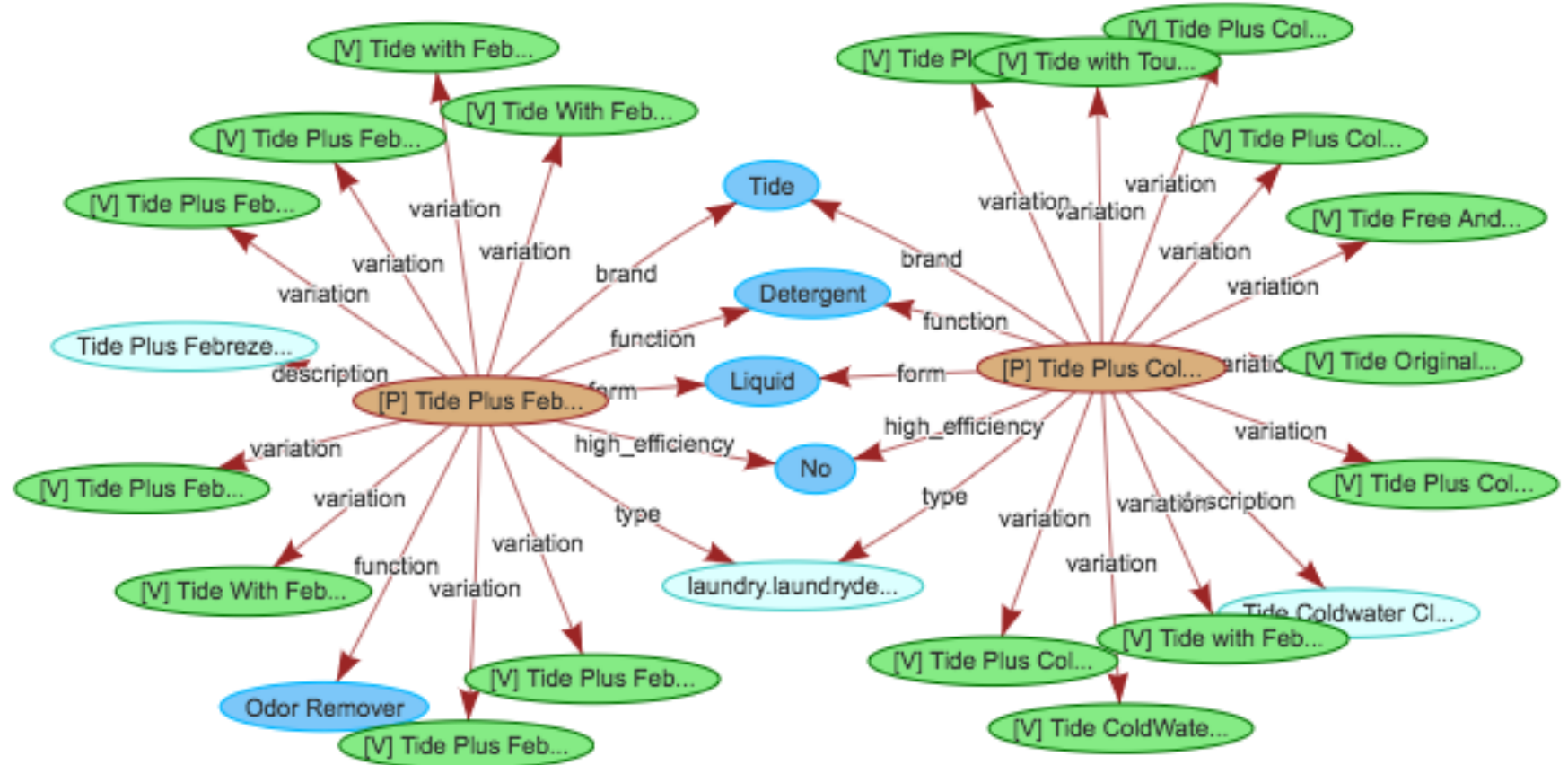
- Changing world: Amazon Product Knowledge Graph

New products

New product categories

New features

Discontinued products



Examples Requiring Change

- Changing requirements: Google Knowledge Graph

An artist must be a person



An artist must be a person OR a vocaloid



Examples Requiring Change

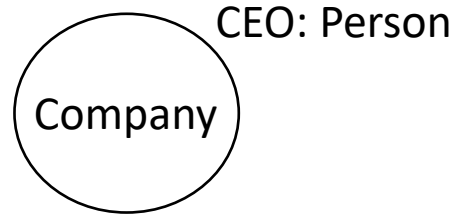
- Changing sources (Google Knowledge Graph)
 - The artists of music albums are obtained from different sources
 - These sources keep changing their data feed
 - The sources used also change over a period of time

Examples Requiring Change

- Changes affecting previous inferences
 - Consider the constraint that a movie theater only shows movies
 - Using this constraint a KG might have previously inferred that certain events are movies
 - More recently the movie theaters are being used for operas, and social events
 - If we had previously derived such events to be movies, we must update them

Examples Requiring Change

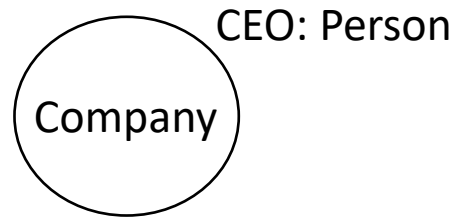
- Changes requiring redesign
 - Initial design: Every company has a CEO. A CEO is represented by name.



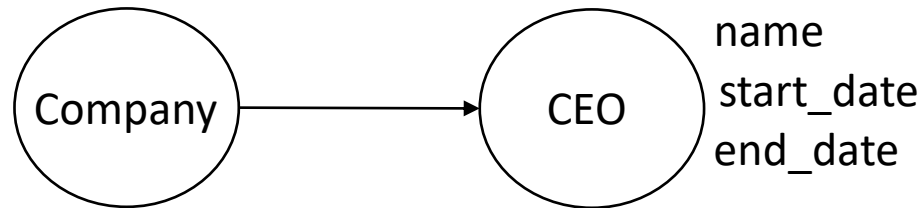
Examples Requiring Change

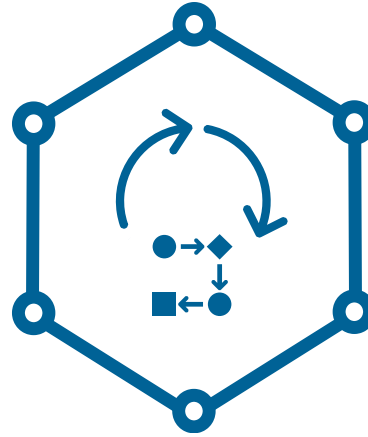
- Changes requiring redesign

- Initial design: Every company has a CEO. A CEO is represented by name.



- Revised design: Every company has a CEO. A CEO is represented by an object that can also record the duration for which the person was a CEO





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